for all these symptoms? It may be due to infection of microbial pathogens or climatic factors or due to mineral deficiency.

In this chapter we are going to learn about classification of minerals, their functions, deficiency diseases and symptoms, nitrogen metabolism and special modes of nutrition. Further, how can these ideas help us to improve productivity in agriculture?

Plants naturally obtain nutrients from atmosphere, water and soil. Carbon, hydrogen and oxygen are called as skeletal elements and constitute about 94% of dry weight. These elements play an important role in the formation of organic compounds such as carbohydrates, fats and protein. These non-mineral elements are obtained from air and water. Minerals are classified based on essentiality. **Arnon** and **Stout** (1939) gave criteria required for essential minerals:

1. Elements necessary for growth and development.
2. They should have direct role in the metabolism of the plant.
3. It cannot be replaced by other elements.
4. Deficiency makes the plants impossible to complete their vegetative and reproductive phase.

12.1 Classification of minerals

12.1.1 Classification of minerals based on their quantity requirements

Essential elements are classified as **Macro-nutrients, Micronutrients** and **Unclassified minerals** based on their requirements. Essential minerals which are required in higher concentration are called Macronutrients. Essential minerals which are required in less concentration called are as Micronutrients.

Minerals like Sodium, Silicon, Cobalt and Selenium are not included in the list of essential nutrients but are required by some plants, these minerals are placed in the list of unclassified minerals. These minerals play specific roles for example, Silicon is essential for pest resistance, prevent water lodging and aids cell wall formation in Equisetaceae (*Equisetum*), Cyperaceae and Gramineae (Table 12. 1).

12.1.2 Classification of minerals based on mobility

If you observe where the deficiency symptoms appear first, you can notice differences in old and younger leaves. It is mainly due to mobility of minerals. Based on this, they are classified into 1. Actively mobile minerals and 2. Relatively immobile minerals (Figure 12.1).

a. Actively mobile minerals

Nitrogen, Phosphorus, Potassium, Magnesium, Chlorine, Sodium, Zinc and Molybdenum.

Deficiency symptoms first appear on old and senescent leaves due to active movement of minerals to younger leaves.

b. Relatively immobile minerals

Calcium, Sulphur, Iron, Boron and Copper shows deficiency symptoms first that appear

Table 12.1: Mineral Types

Macro nutrients	Micro nutrients	Unclassified minerals
Excess than 10 mmole Kg-1 in tissue concentration or 0.1 to 10 mg per gram of dry weight.	Less than 10 mmole Kg-1 in tissue concentration or equal or less than 0.1 mg per gram of dry weight.	Required for some plants in trace amounts and have some specific functions.
Example: C, H, O, N, P, K, Ca, Mg and S	Example: Fe, Mn, Cu, Mo, Zn, B, Cl and Ni	Example: Sodium, Cobalt, Silicon and Selenium

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